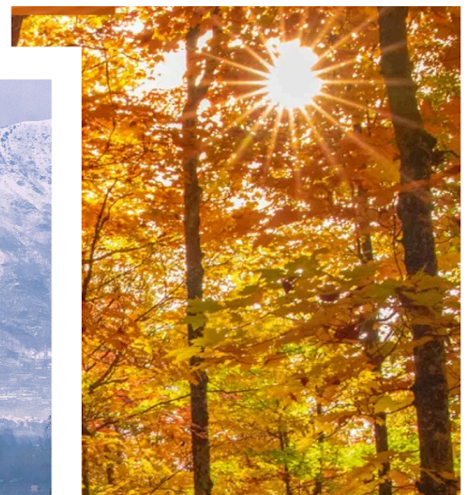


General Knowledge

Indian Geography

STUDY NOTES

Drainage System - Introduction

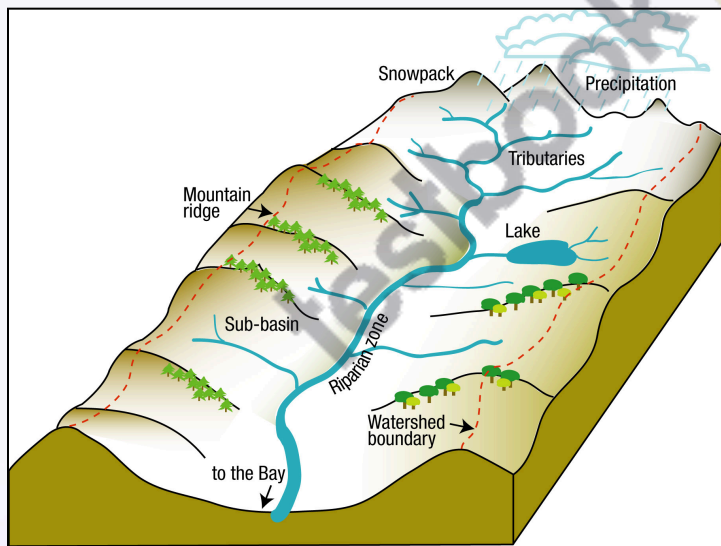


Drainage System: Introduction

Rivers constitute the most useful natural resource, and **India** is blessed with hundreds of large and small rivers that drain the length and breadth of the country. These rivers are a great source of **water** for **irrigation, industry, and domestic purposes** and have attracted the attention of **planners, economists, geographers, geologists, hydrologists, and the like.**

Before learning about the **drainage system** of **India**, it is important to know a few commonly used terms:

- The **flow of water** through well-defined **channels** is known as '**drainage**'; such channels' network is called a '**drainage system**.'
- The **drainage pattern** of an area is the outcome of the **geological period, nature and structure of rocks, topography, slope, amount of water flowing, and the periodicity of the flow.**
- A **river** drains the water collected from a specific area, which is called its '**catchment area**.'
- An area drained by a **river and its tributaries** is called a **drainage basin**.
- The **boundary line** separating one **drainage basin** from the other is known as the **watershed**.
- The **catchments** of large rivers are called **river basins**, while those of small **rivulets** and **rills** are often referred to as **watersheds**.

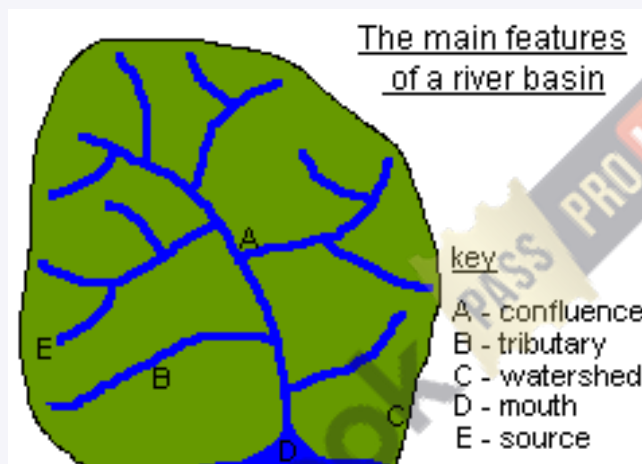


There is, however, a slight difference between a **river basin** and a **watershed**:

- **Watersheds** are **small** in area, while the **basins** cover **larger areas**.
- **River basins** and **watersheds** are marked by **unity**. What happens in one part of the **basin** or **watershed** directly affects the other parts and the unit as a whole. That is why they are accepted as the most appropriate **micro, meso, or macro planning regions**.

River Basin and Watershed

- **River Basin:** A **river basin** is the entire geographical area **drained** by a **river and its tributaries**. It includes all the land where surface water from **rain, melting snow, or ice** converges to a single point, usually the river's **mouth**, where it discharges into another **water body** such as an **ocean, sea, or lake**.
- **Watershed:** A **watershed** is a smaller geographical area that **drains** all its **water** to a common **outlet**, such as a **stream, lake, or river**. It is essentially a **smaller unit** within a **river basin** and is often delineated by **ridges or high points** separating two **drainage areas**.



Major River Basins of India

River Basin	Major River	Length (km)	States Covered	Key Tributaries	Drains Into	Flow Direction
Ganga Basin	Ganga	2,525	Uttarakhand, Uttar Pradesh, Bihar, Jharkhand, West Bengal	Yamuna, Ghaghara, Gandak, Kosi, Son, Damodar	Bay of Bengal	West to East
Brahmaputra Basin	Brahmaputra	2,900 (overall)	Arunachal Pradesh, Assam, Nagaland, Meghalaya	Dibang, Lohit, Subansiri, Manas, Dhansiri	Bay of Bengal	West to East

Godavari Basin	Godavari	1,465	Maharashtra, Telangana, Andhra Pradesh, Chhattisgarh, Odisha	Indravati, Pranahita, Manjira, Sabari, Penganga	Bay of Bengal	West to East
Krishna Basin	Krishna	1,400	Maharashtra, Karnataka, Telangana, Andhra Pradesh	Tungabhadra, Bhima, Malaprabha, Koyna, Musi	Bay of Bengal	West to East
Cauvery Basin	Cauvery	800	Karnataka, Tamil Nadu, Kerala, Puducherry	Kabini, Bhavani, Noyyal, Amaravati, Hemavathi	Bay of Bengal	West to East
Narmada Basin	Narmada	1,312	Madhya Pradesh, Maharashtra, Gujarat	Tawa, Hiran, Banjar, Shakkar, Dudhi, Tendon	Arabian Sea	East to West
Tapi Basin	Tapi	724	Madhya Pradesh, Maharashtra, Gujarat	Purna, Girna, Panjhra, Bori, Waghur, Aner	Arabian Sea	East to West
Mahanadi Basin	Mahanadi	851	Odisha, Chhattisgarh, Jharkhand, Maharashtra	Tel, Ib, Jonk, Hasdeo, Mand, Seonath	Bay of Bengal	West to East
Indus Basin	Indus	3,180 (overall)	Jammu & Kashmir, Himachal Pradesh, Punjab, Haryana, Rajasthan	Jhelum, Chenab, Ravi, Sutlej, Beas	Arabian Sea	North to South (in India)
Penner Basin	Penner	597	Andhra Pradesh, Karnataka	Chitravathi, Papagni, Kunderu	Bay of Bengal	West to East

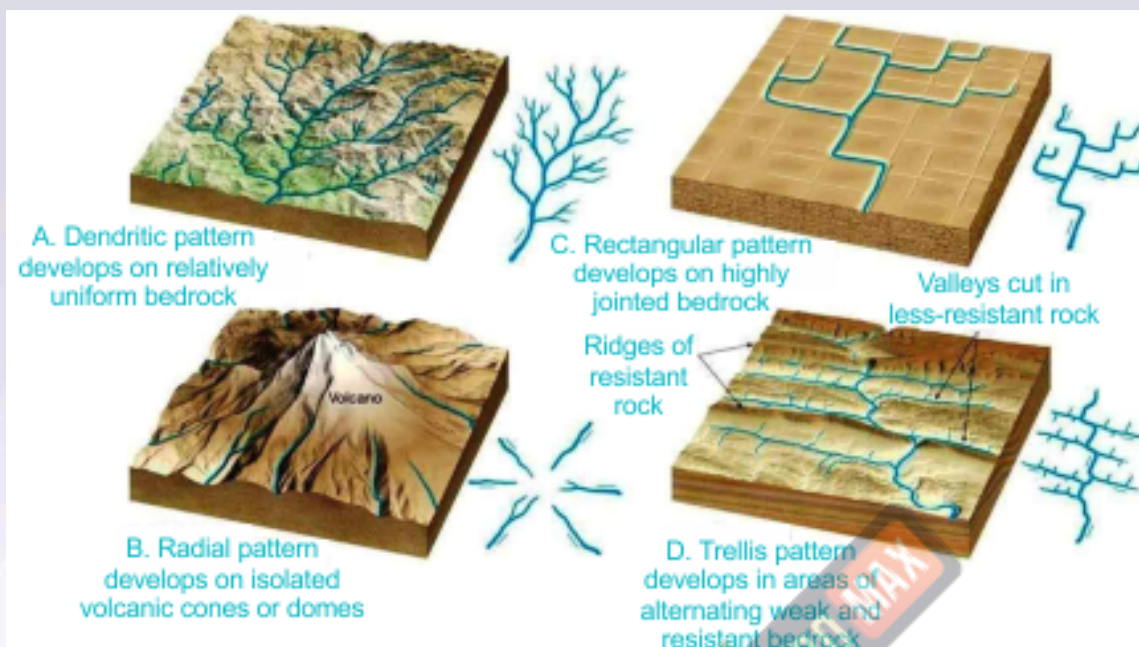
Sabarmati Basin	Sabarmati	371	Rajasthan, Gujarat	Wakal, Sei, Harnav, Hathmati, Khari	Arabian Sea	East to West
Mahi Basin	Mahi	583	Madhya Pradesh, Rajasthan, Gujarat	Som, Anas, Bhadar	Arabian Sea	East to West
Brahmani-Baitarani	Brahmani, Baitarani	799 (combined)	Odisha, Jharkhand	Karo, Sankh, Salandi, Tikira, Kimiria	Bay of Bengal	West to East
Subarnarekha Basin	Subarnarekha	395	Jharkhand, Odisha, West Bengal	Kharkai, Kanchi, Karokari	Bay of Bengal	West to East
Periyar Basin	Periyar	244	Kerala	Muthirapuzha, Cheruthoni, Mullayar	Arabian Sea	East to West

Drainage Patterns

The **streams** within a **drainage basin** form certain **patterns**, depending on the **slope of the land**, **underlying rock structure**, as well as the **climatic conditions** of the area. These are:

- **Dendritic**
- **Trellis**
- **Rectangular**
- **Radial patterns**
- The **dendritic pattern** develops where the **river channel** follows the **slope of the terrain**. The **stream** with its **tributaries** resembles the **branches of a tree**, thus the name **dendritic**.
- A **river** joined by its **tributaries**, at approximately **right angles**, develops a **trellis pattern**. A **trellis drainage pattern** develops where **hard and soft rocks** exist **parallel** to each other.
- A **rectangular drainage pattern** develops on a strongly **jointed rocky terrain**.
- The **radial pattern** develops when **streams flow** in different **directions** from a central **peak** or **dome-like structure**.

A combination of several patterns may be found in the same drainage basin.



Classification of Drainage Systems of India

1 Based on the size of the catchment area:

Division	Size of Catchment Area	Major Rivers/Basins
Major Rivers	> 20,000 sq. km.	Ganga, Brahmaputra, Krishna, Tapi, Narmada, Mahi, Pennar, Sabarmati, Barak, etc.
Medium Rivers	20,000 – 2,000 sq. km.	Kalindi, Periyar, Meghna, etc.
Minor Rivers	≤ 2,000 sq. km.	A good number of rivers, primarily flow in regions of low rainfall.

2 Based on their origin:

- The **Himalayan Rivers:** These are **perennial rivers** and include the **Indus**, the **Ganga**, the **Brahmaputra**, and their **tributaries**.
- The **Peninsular Rivers:** These are **non-perennial rivers** and include the **Mahanadi**, the **Godavari**, the **Krishna**, the **Cauvery**, the **Narmada**, and the **Tapi** and their **tributaries**.

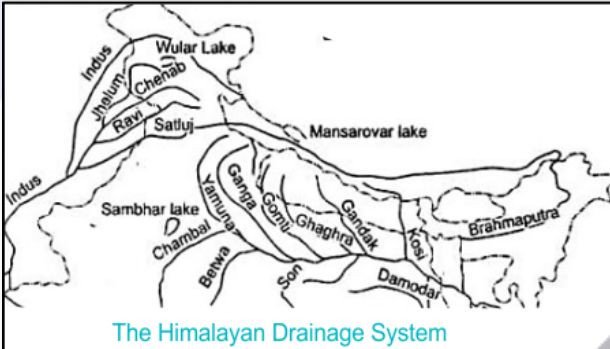
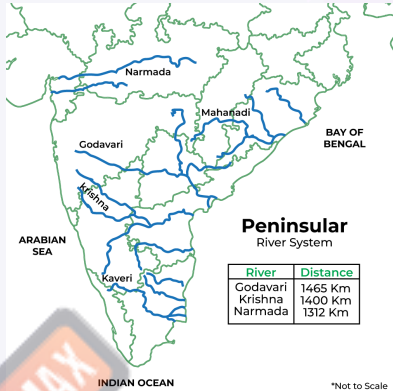
3 Based on Orientation to the Sea:

- The **Bay of Bengal drainage**: Rivers that **drain into the Bay of Bengal** and are **east-flowing rivers**.
- The **Arabian Sea drainage**: Rivers that **drain into the Arabian Sea** and are **west-flowing rivers**.
- They are separated by the **Delhi Ridge**, the **Aravallis**, and the **Sahyadris (water divide)**.

The Bay of Bengal drainage	The Arabian Sea drainage
Rivers that drain into the Bay of Bengal	Rivers that drain into the Arabian Sea
East flowing rivers	West flowing rivers
77 percent of the drainage area of the country is oriented towards the Bay of Bengal	23 percent of the drainage area of the country is oriented toward the Arabian sea
The Ganga, the Brahmaputra, the Mahanadi, the Godavari, the Krishna, the Cauvery, the Penneru, the Pennar, the Vaigai, etc. are some examples.	The Indus, the Narmada, the Tapi, the Sabarmati, the Mahi, and a large number of swift-flowing western coast rivers descending from the Sahyadris are some examples.

The area covered by the Bay of Bengal drainage and Arabian Sea drainage are not proportional to the amount of water that drains through them.

Comparison between the Himalayan and the Peninsular Rivers

Aspects	Himalayan Rivers	Peninsular Rivers
Diagram		
Place of origin	Himalayan mountain covered with glaciers	Peninsular plateau and central highland
Nature of flow	Perennial; receive water from glacier and rainfall	Seasonal; dependent on monsoon rainfall
Type of drainage	Antecedent and consequent leading to dendritic pattern in plains	Super-imposed, rejuvenated resulting in trellis, radial and rectangular patterns
Nature of river	Long course, flowing through the rugged mountains experiencing headward erosion and river capturing; In plains meandering and shifting of course	Smaller, fixed course with well-adjusted valleys
Catchment area	Very large basins	Relatively smaller basin
Age of the river	Young and youthful, active and deepening in the valleys	Old rivers with graded profile, and have almost reached their base levels